

MODULE 7

Business Immersion & Product Thinking

Renaissance Developer Academy — Detailed Lesson Plan

Week 7 of 10 | 5 Days | Full-Time Intensive
Badge Earned: Product-Minded Engineer

Prerequisite: Module 6 — Rapid Builder
DRAFT — March 2026

Module Overview

This is the module that makes most engineers uncomfortable — and it is **the module that separates the Renaissance Developer from a fast coder**. The JD is explicit: this role “shapes technology-driven business strategy, represents the technical perspective at executive level, and is recognized as a bridge between engineering and business.” You cannot bridge what you do not understand.

Module 7 teaches engineers to **think like business insiders**: to read P&L statements, build Lean Canvases, conduct customer discovery interviews, calculate ROI, and write business cases that convince non-technical leaders to invest in technical solutions. Every technical project is reframed through the lens of: **who does this serve, what is the measurable impact, and why should the business care?**

The week prepares students for their capstone project (Modules 9–10) by requiring a Lean Canvas and customer discovery **before any capstone code is written**. This ensures the capstone solves a real problem for a real user — not just a technically interesting exercise.

Connection to the JD: This module maps directly to Business Immersion (“influence domain strategy, exemplify the business immersion mindset”), Problem Discovery (“transform ambiguous situations into clear opportunities”), Stakeholder Management (“manage executive relationships”), and the Polymath behavior (“think like a business insider, translate specialized knowledge into accessible explanations”).

Learning Objectives

1. Create Lean Canvases and value propositions that connect technical solutions to business outcomes
2. Conduct customer discovery interviews using the Jobs-to-be-Done framework and synthesize findings into actionable product insights
3. Read and interpret P&L statements, unit economics, and KPI dashboards to understand what executives care about
4. Write business cases that justify technical investments to non-technical leaders, with ROI projections and risk analysis
5. Use NotebookLM Deep Research for structured industry analysis relevant to your capstone project domain
6. Define the capstone project: problem statement, Lean Canvas, customer discovery findings, and initial technical architecture

Pre-Work (Assigned End of Module 6)

Due before Day 1. Estimated time: 3–4 hours.

1. **Reading:** “The Lean Startup” Chapters 5–8 (Leap, Test, Measure, Pivot or Persevere). If you haven’t read Chapters 1–4, read the full Part One. Upload to NotebookLM and generate an Audio Overview.
2. **Reading:** Gergely Orosz, “The Product-Minded Engineer” (article). Revisit if you read it before Module 2 — it will land differently now that you’ve shipped multiple products.
3. **Capstone Brainstorm:** Come with 3 capstone project ideas. For each: (a) who is the user? (b) what problem does it solve? (c) why do you care about this problem personally? (d) could you find 3–5 real people to interview about it this week?
4. **Financial Literacy Baseline:** Google “how to read a P&L statement” and read one introductory article. Note: what terms are unfamiliar? What feels confusing? Bring your confusion — that’s what Day 2 is for.

Week-at-a-Glance

Day	Theme	Primary Tools	Key Output
Day 1	Lean Canvas & Value Proposition Design	Lean Canvas template, Kiro, NotebookLM	3 Lean Canvases, capstone canvas draft
Day 2	Finance for Engineers & Business Metrics	P&L templates, KPI dashboards, NotebookLM	P&L analysis, ROI calculation, metrics framework
Day 3	Customer Discovery & Jobs-to-be-Done	Interview guides, recording tools	5+ customer interviews, JTBD analysis
Day 4	Business Case Writing & Capstone Definition	All tools, NotebookLM Deep Research	Business case document, capstone project defined
Day 5	Executive Presentations, Reflection & Calibration	All tools, NotebookLM	Business case presentation, development map v7

Day 1: Lean Canvas & Value Proposition Design

Theme: Learn to think about technical solutions as business propositions — who is the customer, what is the value, how does it make money or save money?

Goal: Every student has created 3 Lean Canvases (including one for their capstone idea), understands how to articulate value propositions, and can explain why “who cares?” is the first question a Renaissance Developer asks.

Schedule

09:00	Week 7 Opening: Why Business Thinking Is Not Optional	30 min
	- The uncomfortable truth: most technical projects fail not because the code is bad but because the problem was wrong - Connecting to the Renaissance Developer archetype: — The Strategist Venn Diagram (from Module 1 PPT): Business Strategy + Technical Execution, overlapping as The Strategist — The Polymath mindset: “think like a business insider, translate specialized knowledge into accessible explanations” — The Outcome mindset: “define accountability standards focused on business impact” - This week’s framing: every technical decision is a business decision. The question is never just “does it work?” but “who does it serve and why does it matter?” - Share capstone brainstorm from pre-work: quick round-robin of top ideas (30 sec each)	
09:30	The Lean Canvas Framework (Concept Session)	60 min
	- Why Lean Canvas (not Business Model Canvas): it’s built for uncertainty and speed — perfect for the Renaissance Developer’s “discover then build” approach - The 9 blocks explained through an engineering lens: — Problem: the top 3 pains your user experiences (not features you want to build) — Customer Segments: who are they specifically? (not “everyone” — that’s a red flag) — Unique Value Proposition: one sentence explaining why this matters — the hardest block to fill — Solution: the minimum features that address the top problems (your MVP scope) — Unfair Advantage: what makes this hard to copy? (your domain expertise, unique data, existing relationships) — Revenue Streams: how does this make or save money? (even for internal tools — calculate the cost saved) — Cost Structure: what does it cost to build, run, and maintain? — Key Metrics: the 3–5 numbers that tell you if this is working (not vanity metrics) — Channels: how do users discover and adopt this solution?	

	• Live example: instructor fills a Lean Canvas for a real product (e.g., Slack, Notion, or a Module 6 hackathon prototype) • Common engineer mistakes: starting with Solution instead of Problem, listing features instead of pains, ignoring Revenue
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10:30	Break	15 min
	• Questions about the framework	

10:45	Lean Canvas Workshop: Round 1 — Known Products	60 min
	• EXERCISE 1: Create Lean Canvases for 2 products you use daily • — Pick 2 products: one consumer (e.g., Spotify, Uber, Notion) and one developer tool (e.g., GitHub, Vercel, Datadog) • — Fill every block. Struggling with a block is the learning — don't skip it. • — For Revenue Streams: research how the product actually makes money (it may surprise you) • — For Key Metrics: what would the CEO of this company check every morning? • — For Unfair Advantage: what makes this product hard to replace? (network effects, data moats, switching costs) • Pair review: exchange canvases with a partner. Challenge each block: • — “Is the problem really a top-3 pain? Or is it a nice-to-have?” • — “Is the customer segment specific enough? Could you name 5 real people in this segment?” • — “Is the UVP truly unique? Could I say the same thing about a competitor?” • Use NotebookLM to research products you're less familiar with — upload company blog posts, earnings summaries, or product pages	

11:45	Lunch	60 min
	• Optional: study how great UVPs are written — look at landing pages of products you admire	

12:45	Value Proposition Design Deep Dive	45 min
	• The UVP is the hardest block — and the most important • Value Proposition anatomy: [Outcome] for [Customer] who [Current Pain], unlike [Alternative], our [Solution] [Key Differentiator] • — Example: “Faster incident resolution for on-call engineers who waste 30 min finding the right runbook, unlike searching Confluence, our tool surfaces the relevant runbook automatically based on alert context.” • Testing your UVP: the Mom Test (from Rob Fitzpatrick's book) • — Don't tell people about your idea and ask if they like it (they'll lie to be nice) • — Instead ask about their life: “How do you handle X today? What's the most frustrating part? What have you tried?” • — Facts about their current behavior are worth more than opinions about your solution	

	• Connection to Day 3: customer discovery interviews are the primary tool for testing your UVP
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13:30	Lean Canvas Workshop: Round 2 — Your Capstone Idea	90 min
	• EXERCISE 2: Create a Lean Canvas for your capstone project • — Start with the Problem block: what are the top 3 pains your target user experiences? • — Customer Segments: be specific. Not “developers” but “mid-level backend engineers at companies with 50–500 employees who are on-call at least once a month” • — Solution: what is the absolute minimum you could build in Weeks 9–10 that would validate the value proposition? • — Key Metrics: what would prove this works? (not “users like it” but “users accomplish X in Y% less time”) • — Revenue/Cost: even for a program project, estimate the business value. What would a company pay for this? How much does it cost to build and run? • Structured peer critique: each student presents their canvas to a group of 3–4 (5 min present, 10 min critique) • — The critique group’s job: poke every assumption. “How do you know this is a real problem?” “Why would someone switch from their current workaround?” “What’s your evidence?” • — Document the critique: which assumptions are validated, which are untested, which are risky? • This canvas is REQUIRED before any capstone code is written in Modules 9–10	

15:00	Break	15 min
	• Absorb the feedback	

15:15	Customer Discovery Interview Preparation	45 min
	• EXERCISE 3: Prepare for Day 3’s customer discovery interviews • — Identify 5+ potential interviewees for your capstone problem domain • — Option A: people in your professional network who experience the problem • — Option B: participants sourced through the program’s partner organizations • — Option C: online communities (Reddit, Discord, Slack groups) where potential users congregate • — Write an interview guide using Jobs-to-be-Done framing: • — Background: “Tell me about your role and how [problem domain] fits into your daily work” • — Current behavior: “Walk me through the last time you dealt with [the problem]. What happened?” • — Pain points: “What was the most frustrating part? What do you wish was different?” • — Current solutions: “What have you tried? What worked? What didn’t?”	

	— Value test: “If a tool could [your UVP], how would that change your work? What would it be worth?” — Schedule at least 3 interviews for Day 3 (in-person, video call, or phone) - Do NOT describe your solution in the interview guide — you’re testing the problem, not pitching the product
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16:00	Day 1 Wrap-Up	30 min
	- Quick share: the strongest UVP in the room (peer-nominated) - Commit all canvases to your portfolio repo - HOMEWORK: Schedule your 5 customer discovery interviews for Day 3. Finalize your interview guide. - HOMEWORK: Research your capstone domain using NotebookLM Deep Research: competitors, market size, existing solutions - Preview: tomorrow we learn to speak the language of business — P&L, unit economics, and ROI	

Instructor Notes — Day 1: Day 1 is emotionally charged for many students — business thinking is where engineers feel most out of their depth. The framing matters enormously: this is NOT turning engineers into MBAs. It is giving engineers the vocabulary and frameworks to bridge the gap with business leaders — which is exactly what the JD requires. The Lean Canvas for known products (Exercise 1) builds confidence on familiar ground before the capstone canvas (Exercise 2) forces students into their own uncertainty. The peer critique is the day’s most important activity — push students to be genuinely challenging. A canvas that survives peer critique has a chance of surviving customer contact. Watch for students who fill the Solution block before the Problem block — redirect them. The interview guide preparation (Exercise 3) must happen today so students have time to schedule interviews before Day 3.

Day 2: Finance for Engineers & Business Metrics

Theme: Learn to read, interpret, and use the financial language that executives speak — P&L statements, unit economics, ROI, and KPI dashboards

Goal: Every student can read a simplified P&L, calculate ROI for a technical project, define KPIs for their capstone, and explain why a CFO should care about their proposal.

Schedule

09:00	Finance for Engineers: Why Numbers Matter (Concept Session)	75 min
	- The Renaissance Developer speaks two languages: code and business. Today we learn the second one. - P&L (Profit & Loss) Statement — the single most important financial document: — Revenue: money coming in (from products, services, subscriptions) — COGS (Cost of Goods Sold): direct costs of delivering the product (hosting, support, licenses) — Gross Margin: Revenue minus COGS. This is how much “profit” each unit of product generates. — Operating Expenses (OpEx): salaries, marketing, R&D, office costs — the overhead — EBITDA / Operating Income: gross margin minus OpEx — is the business actually profitable? — Net Income: the bottom line after taxes and interest - Why engineers should care: every technical decision has a P&L impact — Choosing a more expensive cloud service increases COGS and reduces gross margin — Automating a manual process reduces OpEx (headcount or hours saved) — Building a feature that increases retention improves Revenue - Unit Economics — the math of each customer: — CAC (Customer Acquisition Cost): how much does it cost to get one new customer? — LTV (Lifetime Value): how much revenue does one customer generate over their lifetime? — The golden ratio: LTV should be at least 3x CAC for a healthy business — Payback period: how long until CAC is recovered? - KPIs (Key Performance Indicators) — the executive dashboard: — Leading indicators: predict future performance (signup rate, engagement, trial conversion) — Lagging indicators: report past performance (revenue, churn, NPS) — The best KPIs are actionable — if the number moves, you know what to do about it - Live example: walk through a simplified SaaS P&L and explain every line	

10:15	Break	15 min
	Questions about financial concepts	
10:30	P&L Analysis Workshop	75 min
	- EXERCISE 4: Read and analyze a P&L statement - Instructor provides 2 simplified P&L statements from fictional companies: - Company A: a SaaS product with growing revenue but declining margins - Company B: a consulting firm transitioning to a product model - For each P&L, answer: - Is this company healthy? Why or why not? - Where is money being made? Where is it being lost? - If you were the CTO, what technical investment would improve the P&L most? Why? - What KPIs would you track to monitor the health of this business? - EXERCISE 5: Calculate the business impact of a technical project - Scenario: your team proposes automating a manual process that currently takes 4 people 20 hours/week each - Calculate: current annual cost of the manual process (assume loaded salary) - Calculate: estimated build cost (team size × weeks × loaded salary) - Calculate: estimated ongoing cost (hosting, maintenance, monitoring) - Calculate: annual savings and payback period - Calculate: 3-year ROI (Return on Investment) - Write a 1-paragraph executive summary: “We should invest \$X because it will save \$Y over Z years, with a payback period of W months” - Use NotebookLM to research industry benchmarks: typical engineering salaries, hosting costs, maintenance ratios	
11:45	Lunch	60 min
	Optional: find a real P&L from a public company’s annual report and try to read it	
12:45	Metrics Framework for Your Capstone	60 min
	- EXERCISE 6: Define the business metrics for your capstone project - Using your Lean Canvas from Day 1, define: - Primary metric (the ONE number): the single metric that proves your capstone works - Example: “Time to resolve incidents reduced by X%” or “Documentation search time reduced by Y minutes” - Leading indicators (3–5): early signals that predict success - Example: user adoption rate, daily active usage, task completion rate - Guardrail metrics (1–2): things that should NOT get worse as you optimize the primary metric - Example: user satisfaction shouldn’t drop as speed increases; data quality shouldn’t degrade	

	— Business impact metrics: what is the dollar value of improvement? — Example: hours saved × hourly cost = annual value - Document in your capstone planning document alongside the Lean Canvas - Peer review: does your partner's primary metric actually prove the product works? Or is it a vanity metric?
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13:45	Break	15 min
	Stretch	

14:00	Guest Speaker: Business Leader or Forward-Deployed Engineer	60 min
	- EXERCISE 7: Learn from someone who bridges engineering and business daily - Guest speaker session (60 min: 30 min presentation, 30 min Q&A): — Ideal speaker: a forward-deployed engineer, technical product leader, or CTO who came from engineering — Topics to request: how they learned business thinking, how they present technical decisions to executives, what they wish they'd learned earlier — If live speaker unavailable: use a recorded interview or podcast with a technical leader (NotebookLM Deep Dive format) - Students prepare 3 questions each before the session - After the session: 15-min reflection in pairs — what resonated most? What will you apply this week?	

15:00	Industry Analysis with NotebookLM Deep Research	60 min
	- EXERCISE 8: Conduct an industry analysis for your capstone domain - Use NotebookLM's Deep Research feature to compile a cited research report: — Market landscape: who are the existing players solving adjacent problems? — Market size: how big is the potential opportunity? (even rough estimates help) — Trends: what industry trends support your solution? (AI adoption, regulatory changes, workflow shifts) — Competitive analysis: what are 2–3 alternatives your target user could choose instead? — Gaps: what is nobody doing well? Where is the whitespace? - Save the Deep Research report to your NotebookLM notebook and commit a summary to your repo - This analysis feeds into tomorrow's customer discovery interviews (know the landscape before you talk to users)	

16:00	Day 2 Wrap-Up	30 min
	- Quick share: the most surprising thing about reading a P&L (common: "I never realized how thin margins are") - HOMEWORK: Confirm your 5 customer discovery interviews for tomorrow. Review your interview guide one more time.	

	• HOMEWORK: Calculate a rough ROI for your capstone project using today's methods • Preview: tomorrow is customer discovery day — the most important data collection of the program
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Instructor Notes — Day 2: Day 2 is where many engineers discover they've been making P&L-impacting decisions for years without knowing it (choosing cloud providers, setting team sizes, deciding build vs. buy). The P&L walkthrough should use real-world numbers at a scale students can relate to. The ROI calculation exercise (Exercise 5) is deliberately concrete — use actual salary data for the students' market. The guest speaker is a high-impact session if you can secure someone great; a CTO who transitioned from IC to business leadership is ideal. If sourcing a live speaker is difficult, curate a 30-min segment from a relevant podcast or conference talk and structure it as a guided listening session with discussion prompts. The NotebookLM Deep Research exercise prepares students for informed customer discovery interviews — knowing the landscape makes you a better interviewer.

Day 3: Customer Discovery & Jobs-to-be-Done

Theme: Get out of the building — talk to real people who experience the problem you want to solve

Goal: Every student has conducted 5+ customer discovery interviews, synthesized findings using the Jobs-to-be-Done framework, and has data-backed validation (or invalidation) of their capstone problem hypothesis.

Schedule

09:00	Jobs-to-be-Done Framework (Concept Session)	45 min
	- The core insight: customers don't buy products — they “hire” products to do a job in their life - JTBD anatomy: — When [situation], I want to [motivation], so I can [expected outcome] — Example: “When I get paged at 3 AM, I want to quickly find the relevant runbook, so I can resolve the incident without waking up the whole team” - The four forces of progress (why people switch to new solutions): — Push: dissatisfaction with the current solution (“the spreadsheet is too slow”) — Pull: attraction of the new solution (“this tool looks much faster”) — Habit: attachment to the current way (“but I know how the spreadsheet works”) — Anxiety: fear of the new (“what if this tool loses my data?”) - For adoption to happen: Push + Pull must be greater than Habit + Anxiety - Connection to Module 6's user testing: there you tested a solution. Today you test the PROBLEM. - Interview tips refresher: ask about behavior, not opinions. Listen for emotions. Follow the energy.	
09:45	Interview Practice: Mock Sessions	45 min
	- EXERCISE 9: Practice customer discovery interviews in pairs — Person A plays the interviewer (using their real interview guide) — Person B plays a user who experiences the problem (roleplay based on A's problem domain) — 10 min interview + 5 min feedback: “You asked a leading question here” “You missed following up on that emotion” — Switch roles and repeat - Common mistakes to practice avoiding: — Leading questions: “Don't you think X would be better?” → “How do you handle X today?” — Pitching: “We're building a tool that...” → “Tell me about the last time you...” — Surface-level answers: “It's fine” → “Can you walk me through specifically what happened last Tuesday?” — Confirmation bias: only hearing what supports your hypothesis. Listen for disconfirming evidence.	

10:30	Break	15 min
	Final interview prep	
10:45	Customer Discovery Interviews: Session Block 1	90 min
	- EXERCISE 10: Conduct real customer discovery interviews - Each student conducts 2–3 interviews during this block: - In-person interviews (if participants are local): find a quiet room, 20–25 min each - Video call interviews: Zoom/Meet/Teams, camera on, 20–25 min each - Take notes in real-time or record (with permission) and transcribe later - For each interview, document: - Interviewee profile: role, company size, relevant context (anonymized) - Current behavior: how they handle the problem today - Pain intensity: on a scale of 1–10, how frustrating is this problem? (behavior matters more than the number) - Key quotes: exact words that reveal emotion, frustration, or workarounds - Jobs-to-be-Done: what “job” is the user trying to accomplish? What are they “hiring” their current tool to do? - Willingness to switch: would they try something new? What would it take? - Surprise: what did you learn that you didn’t expect?	
12:15	Lunch	45 min
	Quick lunch — more interviews ahead	
13:00	Customer Discovery Interviews: Session Block 2	90 min
	- EXERCISE 10 (continued): Complete remaining interviews (2–3 more) - Target: 5+ total interviews by end of this block - Adjust your questions between interviews based on what you’re learning - If everyone mentions the same pain, probe deeper: “Why hasn’t this been solved already?” - If nobody mentions your hypothesized pain, explore: “What IS the biggest frustration in your workflow?” - The interview is working if you’re learning things that surprise you - The interview is NOT working if every interviewee confirms your hypothesis perfectly (that’s confirmation bias)	
14:30	Break	15 min
	Decompress after interviews	
14:45	Customer Discovery Synthesis Workshop	75 min

	• EXERCISE 11: Synthesize your interview findings • Use the Jobs-to-be-Done framework to structure your synthesis: • — JTBD Map: what are the top 3 jobs your interviewees are trying to accomplish? • — Pain Ranking: rank the pains by frequency (how many interviewees mentioned it) and intensity (how strongly they felt it) • — Behavior Patterns: what workarounds are people using? (workarounds reveal real pain — people don't build workarounds for minor annoyances) • — Four Forces Analysis: for each interviewee, what is the push, pull, habit, and anxiety? • — Quotes Board: the 5 most powerful quotes from your interviews (these go in your business case) • Validate or invalidate your Lean Canvas assumptions: • — Green: validated by 3+ interviews with strong evidence • — Yellow: partially validated, needs more data • — Red: invalidated — the assumption was wrong • — If your core problem hypothesis was invalidated: that's a successful discovery, not a failure. Pivot your capstone. • Write a Customer Discovery Report (committed to repo): • — Methodology: interview count, profile mix, interview format • — Key findings: top 3 jobs, top 3 pains, top 3 workarounds • — JTBD statements: for each major finding • — Evidence: quotes and behavioral patterns • — Lean Canvas update: which blocks changed based on discovery?
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16:00	Day 3 Wrap-Up	30 min
	• Quick share: "The most surprising thing I learned from my interviews was..." • Pivot check: did anyone's capstone direction change based on discovery? (celebrate this — it means the process works) • HOMEWORK: Finish your Customer Discovery Report. Update your Lean Canvas based on findings. • HOMEWORK: Start drafting your business case for tomorrow • Preview: tomorrow we write business cases and formally define the capstone project	

Instructor Notes — Day 3: Day 3 is the most logistically complex day of Module 7. Interview sourcing must begin weeks in advance — if students can't find 5 interviewees on their own, the program should provide a pool of 15–20 participants from partner organizations (this is worth the investment). Remote interviews via video call work well for students whose target users aren't local. The most common Day 3 failure: students who pitch their solution during the interview instead of discovering the problem. If you observe this in the mock sessions (Exercise 9), coach hard — it's the difference between validated learning and self-reinforcing delusion. The synthesis workshop is where insights crystallize — push students to identify patterns across interviews, not just summarize individual conversations. Students whose core hypothesis is invalidated need special encouragement: this is the Lean Startup working as designed.

Day 4: Business Case Writing & Capstone Definition

Theme: Transform discovery insights into a compelling business case and formally define your capstone project with business justification

Goal: Every student has a written business case document, a finalized capstone Lean Canvas, a customer discovery report, and a preliminary technical architecture for Weeks 9–10.

Schedule

09:00	The Business Case: Concept & Structure (Concept Session)	45 min
	• What a business case IS: a structured argument for why a business should invest resources in a technical initiative • What a business case IS NOT: a technical spec (that comes later), a feature list, or a sales pitch • The anatomy of a compelling business case: • — Section 1 — Problem Statement: the pain, supported by customer discovery evidence • — Section 2 — Proposed Solution: what you will build, at the user benefit level (not the technical implementation level) • — Section 3 — Market Context: industry landscape, competitors, market gap (from NotebookLM research) • — Section 4 — Business Impact: ROI calculation, hours/cost saved, revenue potential, key metrics • — Section 5 — Investment Required: team size, timeline, technology costs, ongoing maintenance • — Section 6 — Risks & Mitigations: what could go wrong and how you'll manage it • — Section 7 — Recommendation: clear ask with a timeline • Writing for the audience: executives read differently than engineers • — Lead with the conclusion ("We should invest \$X because..."), not the background • — Use numbers, not adjectives ("30% reduction" not "significant improvement") • — One page if possible. Two pages maximum. Appendix for supporting details. • — The Pyramid Principle: state the answer first, then provide the supporting arguments	
09:45	Business Case Writing Workshop	105 min
	• EXERCISE 12: Write a business case for your capstone project • Using the structure from the concept session, draft each section: • — Problem Statement: ground it in customer discovery quotes and behavioral data • — Proposed Solution: describe what the user experiences, not what the code does	

	• — Market Context: use your NotebookLM Deep Research findings • — Business Impact: ROI calculation from Day 2's methods (specific numbers, even if estimated) • — Investment Required: realistic estimate for a 2-week capstone build • — Risks: at least 3 risks with specific mitigations • — Recommendation: a clear, one-sentence ask • Use Claude Code CLI or Kiro to help draft — but this is about YOUR judgment, not AI generation • — AI can help with formatting, financial calculations, and competitive analysis • — AI CANNOT replace your customer discovery insights, your ROI assumptions, or your judgment about what matters • Target: 1–2 pages of polished business writing. Concise. Evidence-based. Audience-appropriate.
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11:30	Break	15 min
	• Stretch	

11:45	Peer Review: Business Case Critique	45 min
	• EXERCISE 13: Exchange business cases with a peer for structured critique • Reviewer reads the business case and answers: • — “After reading this, would I fund this project? Why or why not?” • — “What is the strongest argument? What is the weakest?” • — “Is the ROI calculation credible? Are the assumptions reasonable?” • — “Is this written for an executive or for an engineer? (it should be for an executive)” • — “What question would a skeptical CFO ask that this business case doesn’t answer?” • Written feedback on the document (tracked changes or comments) — not just verbal • Revise based on feedback	

12:30	Lunch	60 min
	• Optional: continue revising	

13:30	Capstone Project Definition	90 min
	• EXERCISE 14: Formally define your capstone project (Weeks 9–10) • The capstone definition document includes all work from this week, organized as: • — Lean Canvas (final version, updated with customer discovery findings) • — Customer Discovery Report (from Day 3 Exercise 11) • — Business Case (from today’s Exercise 12, revised after peer feedback) • — Metrics Framework (from Day 2 Exercise 6) • — Preliminary Technical Architecture:	

	— High-level system design (use Module 4’s 5-step framework, but keep it light — this is a sketch) — Tech stack choice with brief ADR — Key user stories (5–8 that define the MVP scope for 2 weeks) — Deployment plan (reuse Module 3–5 patterns: Docker, GitHub Actions, cloud-agnostic) — Industry Analysis (from Day 2 Exercise 8) - This document is the “gate” for entering the capstone — it must be approved before Week 9 coding begins - Use Kiro to generate initial technical specs from the capstone definition — then enhance with your own architecture judgment
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15:00	Break	15 min
	Stretch	

15:15	Presentation Preparation	45 min
	- EXERCISE 15: Prepare an executive-style presentation of your business case - Tomorrow’s presentation format (5 minutes per student): — The Problem (1 min): user pain, supported by discovery quotes — The Opportunity (1 min): market context, gap, your unique insight — The Proposal (1 min): what you’ll build, for whom, with what expected impact (metrics) — The Ask (1 min): investment required, ROI, timeline, risks — Why Me (1 min): why are you the right person to build this? (your domain insight, technical skills, customer relationships) - This is practice for the JD’s requirement: “represent the technical perspective at executive level” - Rehearse once with a partner. Time it. Get feedback. - Key: NO live demo, NO technical architecture slides. This is a BUSINESS presentation.	

16:00	Day 4 Wrap-Up	30 min
	- Commit all capstone definition documents to portfolio repo - HOMEWORK: Rehearse your presentation one more time. Refine the business case based on any remaining peer feedback. - Preview: tomorrow — executive presentations and the formal capstone approval	

Instructor Notes — Day 4: Day 4 is where Module 7 becomes tangible. The business case writing exercise is the week’s most challenging deliverable for most engineers. The most common failures: too technical (“we’ll use React and PostgreSQL” in the Solution section), too vague (“it will save time” without numbers), or too long (engineer’s instinct is to include every detail). Coach toward: specific, quantified, concise, audience-appropriate. The peer critique (Exercise 13) often reveals that engineers write business cases for other engineers, not for executives — the question “Is this written for an executive?” usually triggers a rewrite. The

capstone definition (Exercise 14) is the formal bridge between Module 7 and Modules 9–10. If a student's capstone idea is weak (vague problem, no customer discovery evidence, no business justification), this is the time to course-correct — not Week 9.

Day 5: Executive Presentations, Reflection & Calibration

Theme: Present your business case to the cohort (simulating an executive audience), receive structured feedback, and prepare for the final stretch of the program

Goal: Every student has delivered a business presentation, has received capstone approval, and has updated their Personal Development Map to reflect the business immersion growth.

Schedule

09:00	Executive Presentation: Business Case Showcase	120 min
	• Every student presents: 5 minutes each, strictly timed • The audience plays the role of an investment committee (executives evaluating proposals): • — Problem clarity: is the pain real and well-evidenced? • — Solution viability: is the proposed solution realistic for 2 weeks? • — Business impact: are the ROI numbers credible? • — Customer evidence: was the discovery rigorous? Are the findings convincing? • — Presentation quality: was it clear, concise, and executive-appropriate? • After each presentation: 2 minutes of “executive Q&A” (tough but fair questions): • — “What’s your biggest risk and how will you know if it’s materializing?” • — “Why should we invest in this over the other proposals?” • — “Your ROI assumes X — what if that assumption is wrong by 50%?” • Scoring (1–5 per criterion) by all audience members • Awards: Best Business Case, Strongest Customer Discovery, Most Compelling Presentation, Best ROI Analysis	
11:00	Break	15 min
	• Decompress after presentations	
11:15	Capstone Approval & Feedback	45 min
	• EXERCISE 16: Capstone approval session • Instructor reviews each student’s capstone definition and provides: • — Approved: capstone is well-defined, evidence-based, and achievable in 2 weeks • — Approved with conditions: specific items to address before Week 9 (scope narrowing, more discovery, technical feasibility check) • — Needs revision: fundamental issues that require rethinking (rare by this point if Modules 6–7 worked well) • Students with conditions have the weekend and Module 8 to address them	

	Peer matching: connect students whose capstones share themes or technologies for mutual support in Weeks 9–10	
12:00	Lunch	60 min
	Celebrate — you just pitched a business case like a startup CTO	
13:00	Week 7 Retrospective & Development Map Update	60 min
	- Sailboat retrospective: — Wind: What worked? (often: customer discovery was eye-opening, Lean Canvas forced clarity) — Anchor: What was hard? (often: financial calculations, writing for executives, resisting urge to lead with technology) — Rocks: Concerns about Module 8 (communication/leadership) and capstone (Modules 9–10)? — Island: What are you excited about for the capstone? - EXERCISE 17: Personal Development Map v7 — Re-rate on five mindsets: how has Outcome orientation grown? (SLOs, business metrics, ROI thinking) — How has Polymath thinking expanded? (you now speak P&L, JTBD, and UVP as well as API, CI/CD, and SLO) — Competency layers: Module 7 targeted Layer 4 (Strategist — Business Immersion, Problem Discovery, Stakeholder Management) — The Strategist Venn: re-plot yourself. Has the overlap between Business Strategy and Technical Execution grown? — Capstone readiness: are you ready to build something that matters? - Instructor 1:1s through the afternoon	
14:00	Impossible Stuff Day: Self-Directed Exploration	105 min
	- Week 7 Impossible Stuff Day suggestions: — Conduct 2–3 more customer discovery interviews to strengthen your capstone evidence — Build a financial model for your capstone: spreadsheet with revenue projections, cost estimates, break-even analysis — Research competitive products in depth — sign up for trials, analyze their onboarding, study their pricing — Draft a landing page for your capstone (even before the product exists — test the UVP) — Read “Inspired” by Marty Cagan Chapters 1–5 for deeper product thinking — Start reading the full OpenClaw monorepo architecture in preparation for Module 8 — Write a blog post: “What I learned about business by building software” - Share explorations in last 15 min	
15:45	Break	15 min

	• Stretch
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16:00	Badge Ceremony & Week 7 Close	30 min
	• Badge ceremony: Product-Minded Engineer badge for students who completed all deliverables • Achievements: best business case, strongest customer discovery, most compelling presentation, best industry analysis • Read the Developer Manifesto together (recurring ritual) • 70% through the program. You can now build (M3), design (M4), ensure reliability (M5), deliver under pressure (M6), AND think like a business insider (M7). • Module 8 preview: Communication, Leadership & Enterprise Scale — the final pillar before capstone — Multi-audience communication, OpenClaw monorepo analysis, security audit, leadership simulation • — This is where you prove you can operate at the organizational level the JD demands	

Instructor Notes — Day 5: The executive presentations are the week’s culminating experience and should feel like a real investment pitch. Tough but fair Q&A is essential — softball questions don’t prepare students for real executive interactions. Coach students who get flustered by Q&A to pause, think, and answer honestly (“I don’t have data on that yet, but here’s how I’d find out” is a strong answer). The capstone approval session is a formal gate — take it seriously. Students whose capstones are weak need honest feedback now, not encouragement to proceed with a flawed plan. The Development Map v7 should show dramatic growth in the Outcome and Polymath dimensions — if it doesn’t, probe whether the student engaged fully with the business exercises or treated them as homework to get through.

Appendix A: Exercise Summary

#	Day	Exercise	Type
1	1	Lean Canvas for 2 known products (consumer + developer tool) with peer critique	Framework
2	1	Lean Canvas for capstone project with structured peer critique and assumption mapping	Framework
3	1	Customer discovery interview guide preparation + schedule 5 interviews	Preparation
4	2	P&L analysis: read 2 fictional P&Ls, identify health indicators, propose CTO-level investments	Analysis
5	2	ROI calculation: quantify business impact of automating a manual process	Financial
6	2	Define business metrics for capstone: primary metric, leading indicators, guardrails, dollar impact	Framework
7	2	Guest speaker session: business leader or forward-deployed engineer (30 min + 30 min Q&A)	Learning
8	2	NotebookLM Deep Research: industry analysis for capstone domain (competitors, market, trends, gaps)	Research
9	3	Mock customer discovery interviews in pairs with feedback	Practice
10	3	Conduct 5+ real customer discovery interviews with structured documentation	Customer Research
11	3	Synthesize findings: JTBD map, pain ranking, four forces, quotes board, Lean Canvas update	Synthesis
12	4	Write business case: 7-section document (problem, solution, market, impact, investment, risks, recommendation)	Business Writing
13	4	Peer review business case: structured critique with tracked changes	Collaborative
14	4	Capstone project definition: Lean Canvas + discovery + business case + metrics + technical architecture	Planning
15	4	Prepare executive-style presentation: 5-min business pitch (no live demo, no technical slides)	Presentation
16	5	Capstone approval: instructor review and feedback	Assessment
17	5	Personal Development Map v7: Outcome + Polymath growth, Strategist Venn re-plot	Reflection

Appendix B: Deliverables Checklist

Complete all items to earn the Product-Minded Engineer badge:

1. **Lean Canvases** — 3 canvases total: 2 for known products (practice) and 1 for capstone project (final version updated with customer discovery), all committed to portfolio repo
2. **Customer Discovery Report** — 5+ interview syntheses with JTBD analysis, pain ranking, four forces analysis, key quotes, and Lean Canvas validation/invalidation mapping
3. **Business Case Document** — 7-section business case (1–2 pages): problem, solution, market, impact with ROI calculation, investment, risks, recommendation. Peer-reviewed with tracked changes.
4. **Business Metrics Framework** — Primary metric, leading indicators, guardrail metrics, and dollar impact calculation for capstone project
5. **Industry Analysis** — NotebookLM Deep Research report: competitors, market size, trends, competitive gaps, committed as summary in portfolio repo
6. **Capstone Project Definition** — Complete document including: final Lean Canvas, customer discovery report, business case, metrics framework, preliminary technical architecture with ADR and user stories — approved by instructor
7. **Executive Presentation** — 5-minute business case presentation delivered within time limit, with executive-style Q&A handled
8. **Personal Development Map v7** — Updated with Outcome and Polymath growth, Strategist Venn re-plotted, capstone readiness assessment

End of Module 7 Lesson Plan